

scripts/scan-no-hardcoded-ipv4.sh

— User Guide

Last verified: 2026-07-01 (synced to the §6.R slirp 10.0.2.0/24 emulator-helper exemption added this cycle; prior: 2026-05-15 §6.AD-debt task #61 backfill) **Inheritance:** HelixConstitution §11.4.18 (script documentation mandate)

Overview

Stub doc generated from the script's in-source header comment. See scripts/scan-no-hardcoded-ipv4.sh for canonical behavior. This stub exists so the CM-SCRIPT-DOCS-SYNC pre-push gate (.github.com/pre-push Check 9) starts gating modifications to this script.

The script's own header documentation (verbatim):

scripts/scan-no-hardcoded-ipv4.sh — standalone §6.R IPv4 scanner.

Purpose: enforce the §6.R No-Hardcoding Mandate clause that no IPv4 literals appear in tracked source outside the exemption set. §4.5.10 (CONTINUATION.md): the IPv4 enforcement is staged — this scanner is the mechanical gate that lands the rule.

Exit codes:

- 0 — no IPv4 violations
- 1 — IPv4 violation(s) found (paths printed to stderr)

Exemptions (kept in lockstep with the §6.R clause body):

.env.example	— placeholder file
.lava-ci-evidence/	— forensic anchors +
matrix evidence	
docs/**/*.*md	— design docs,
plans, incident notes	
submodules/	— submodules
vendored at pinned hash	
*_test.go, *Test.kt, *Tests.kt, *Test.java	— synthetic test
fixtures	
src/test/, src/androidTest/	— test source roots
fixtures/	— test HTML/JSON
fixtures	
CHANGELOG.md	— release notes may

reference IPs in incident summaries
*.md, *.json, *.xml, *.yaml, *.yaml – external config +
docs are legitimate home for
connection literals (Android
network_security_config.xml whitelists LAN
provisioning .yaml lists service ranges, Grafana
Code that READS these files endpoints, etc.).
is what §6.R
targets.

Loopback / docs-prefix IPs that are universally permitted (RFC
5737,
documentation-only) are filtered AFTER the file-level exemption so
a
code file that legitimately uses 127.0.0.1 or 0.0.0.0 still passes
–
those are not "hardcoded connection addresses" in the §6.R sense
(they cannot connect to anything but the local host or the
wildcard).

Narrow path+range exemption – Android-emulator QEMU user-mode
(slirp)
constants in the 10.0.2.0/24 range inside autonomous-qa emulator
helper
scripts (scripts/autonomous-qa/*emulator*.sh). Within that /24 the
helper
uses host .2 (slirp gateway / host-loopback alias), DNS .3 (the
emulator's
built-in DNS server), and guest .15 (the guest NIC address) –
platform-FIXED
by the Android emulator slirp stack, not configurable, cannot
drift. They are
the emulator equivalent of 127.0.0.1, not a deployment address.
The exemption
is BOTH path-scoped (the autonomous-qa emulator helper) AND range-
scoped
(10.0.2.0/24): a routable IP in those same files, or any 10.0.2.x
literal in
any other tracked file, is still flagged. Line-removal filter,
identical in
mechanism to the reserved-address filter above.

Usage

See the script's in-source comment block (above) for canonical usage examples.

Maintenance

When this script is modified, update this document in the same commit (CM-SCRIPT-DOCS-SYNC requires it). Per §11.4.18, the documentation **MUST** stay in sync with the codebase — no doc may be out of sync with its script.

Cross-references

- `scripts/scan-no-hardcoded-ipv4.sh` — the script itself
- `docs/helix-constitution-gates.md` — gate inventory
- HelixConstitution Constitution.md §11.4.18 (the mandate)
- Lava CLAUDE.md §6.AD (HelixConstitution Inheritance)